

Bilal Amin

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EDUCATION

Universität des Saarlandes <i>M.Sc. Data Science and Artificial Intelligence</i> <i>Grade: 1.9 (German grading system)</i>	Apr. 2024 – Present Saarbrücken, Germany
University of Bergen (Erasmus+ Exchange Semester) <i>Exchange Student, Computer Science</i>	Jan. 2026 – Jun. 2026 Bergen, Norway
National University of Sciences & Technology <i>Bachelor's in Mechatronics Engineering (Computer Science)</i> <i>Grade: 1.7 (German grading system)</i>	Sep. 2017 – Sep. 2021 Islamabad, Pakistan

EXPERIENCE

Working Student – Internet Architecture Group <i>Max Planck Institute for Informatics (MPI)</i> - Developed an automated Python-based evaluation framework for comparing congestion control algorithms in NS-3 across varied network conditions. - Designed system tests and quantitative metrics to evaluate network performance across simulation scenarios. - Built a dashboard to visualize performance metrics and support comparison of algorithm behavior. - Analyzed simulation results to identify bottlenecks, optimization opportunities, and design trade-offs.	Oct. 2024 – Mar. 2025 Saarbrücken, Germany
Data Analyst <i>Convergent Business Technologies</i> - Developed reproducible Python and SQL workflows for cleaning, transforming, and analyzing structured datasets. - Created analytical dashboards and reports to communicate patterns, trends, and data-driven insights.	Oct. 2023 – Mar. 2024 Islamabad, Pakistan
Web Development Intern <i>CureMD</i> - Designed user interfaces using HTML, CSS, and JavaScript, improving usability and accessibility for end users. - Developed APIs in the .NET framework, enhancing data integration and tool development.	Jun. 2023 – Aug. 2023 Lahore, Pakistan
Technical Intern <i>STEP (Space Technology Education and Popularization)</i> - Led workshops teaching programming fundamentals using Scratch, focusing on hands-on learning for underprivileged youth. - Created educational materials to simplify complex technology concepts, fostering interest in STEM fields.	Aug. 2021 – Oct. 2021 Islamabad, Pakistan

PROJECTS

GermanLyft – AI-Powered German Learning Platform - Developed a full-stack German-learning platform using React, TypeScript, Flask, and the Groq LLM API, targeting primarily CEFR A1–A2 learners. - Built an interactive AI tutor for grammar explanations, sentence correction, vocabulary practice, conversational learning, and dynamically generated exercises. - Implemented placement tests, quizzes, user accounts, profiles, and Anki .apkg deck importing, with a REST-based Vite/React and Flask architecture deployed on Vercel and Render.	2026
Smart StudyDesk: AI-Powered Study Organizer - Designed a cross-platform study application for organizing, summarizing, and reviewing lecture materials, with AI-based summaries, personalized quizzes, progress tracking, and weak-area analysis. - Conducted usability testing using semi-structured interviews and NASA-TLX workload analysis, then improved navigation, folder organization, and feedback visibility based on user feedback.	2025
AI Learning Assistant using LLMs (RAG) Built an AI-powered learning assistant using Python, LangChain, Streamlit, and FastAPI to support	2025

document-based learning.

- Integrated a locally hosted Llama model with RAG, vector embeddings, ChromaDB retrieval, and citation-based responses for grounded question answering from uploaded PDFs.
- Developed automatic generation of summaries, quizzes, and study materials from uploaded course documents.

MarketPulse Website Design and HCI Analysis

2024

- Designed and developed a responsive website for MarketPulse using HTML, CSS, and JavaScript.
- Applied HCI principles including consistency, simplicity, feedback, visibility, constraints, and accessibility to improve information structure and user interaction.

Side Hustle Launchpad – Tech4Europe Hackathon, KIT

2025

- Developed a full-stack web application during the Tech4Europe Hackathon at Karlsruhe Institute of Technology, Germany.
- Built analytics dashboards and personalized recommendations for small-business planning using FastAPI and MongoDB for backend development and data management.

MOOCs AND RELEVANT COURSEWORK

- **Saarland University:** Data Networks, Neural Networks: Theory and Implementation, Elements of Machine Learning, High-Level Computer Vision, Generative AI, Attacks Against Machine Learning Models, and Competitive Programming.
- **University of Bergen, 2026:** Qualitative Analysis and Functional Programming.
- **Certifications:** Machine Learning Specialization – DeepLearning.AI and Stanford Online; CS50's Introduction to Artificial Intelligence with Python – Harvard University; Google Data Analytics Professional Certificate – Google.

TECHNICAL SKILLS AND INTERESTS

Programming: Python, JavaScript, TypeScript, SQL, HTML, CSS, C++, C#, Java, Dart

Machine Learning & Data Science: Scikit-Learn, PyTorch, TensorFlow, Keras, Hugging Face Transformers, Pandas, NumPy, PySpark, Data Preprocessing, Statistical Analysis, Regression Modeling, Model Evaluation, SHAP, Matplotlib

Generative & Agentic AI: Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Prompt Engineering, LangChain, LangGraph, Multi-Agent Systems, Groq API, Ollama, ChromaDB

Web & App Development: Flask, FastAPI, ReactJS, Vite, Streamlit, Flutter, REST APIs

Databases: MySQL, MongoDB, SQLite, Supabase

Cloud, MLOps & Dev Tools: Microsoft Azure, MLflow, Docker, Git, GitHub, GitHub Actions, CI/CD, Vercel, Render, VS Code, Jupyter

Visualization & Design: Power BI, Tableau, Figma, Canva

Interests: Artificial Intelligence, Applied Machine Learning, LLM Applications, Agentic AI, Generative AI, Intelligent Learning Systems, Data Engineering, Data Visualization